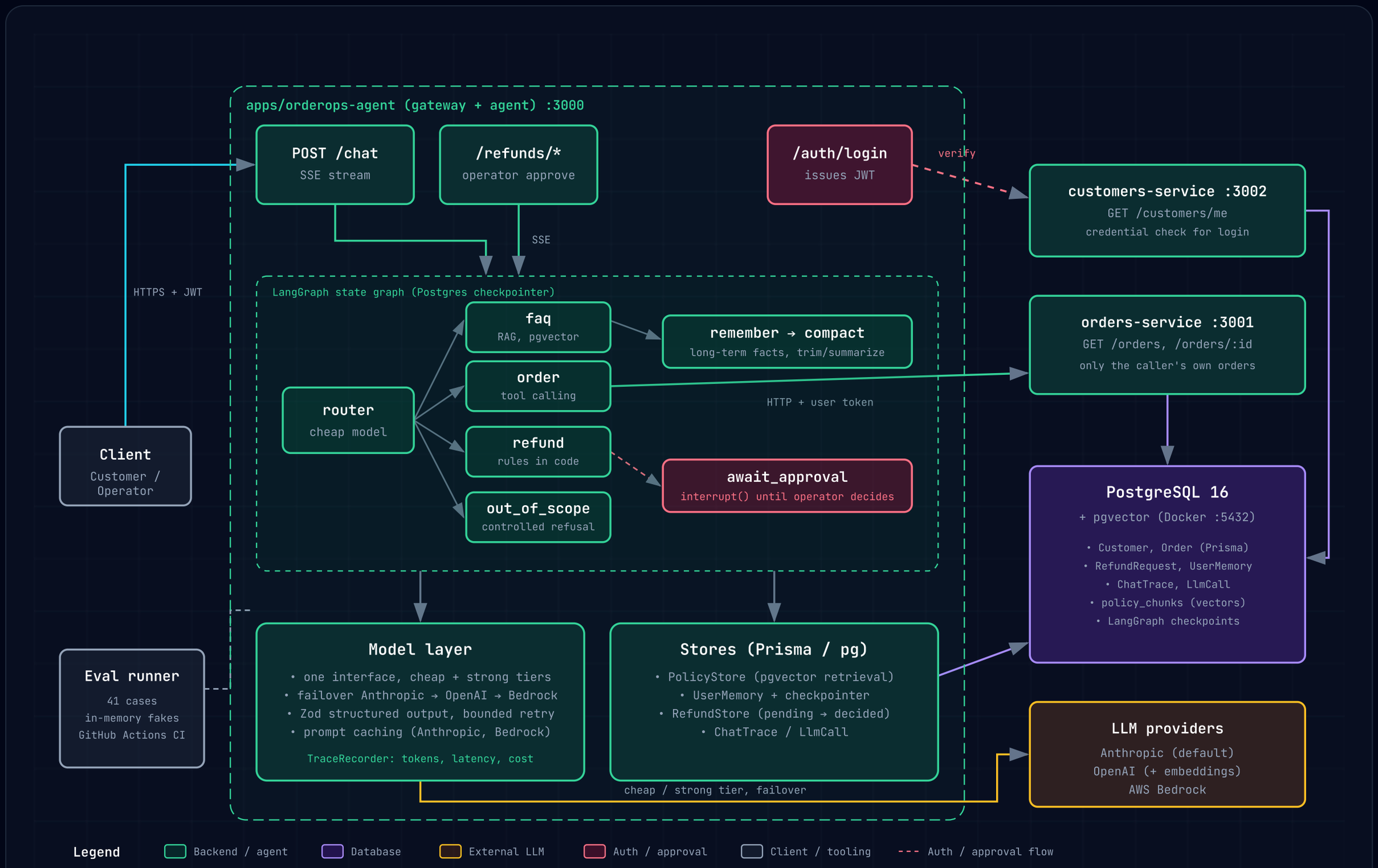
OrderOps Agent Architecture

NestJS gateway + LangGraph agent, two mock microservices, Postgres + pgvector, three LLM providers



Request flow

- Client logs in at /auth/login, gateway verifies with customers-service and signs a JWT
- POST /chat streams SSE: token, tool_call, approval_required, final, error
- router picks faq, order, refund or out_of_scope
- order tools call orders-service with the caller's token

Guardrails

- Refund rules enforced in code: delivered, 30 days, amount ≤ total, one open request
- Zod validation with bounded retry and deterministic fallback
- interrupt() pauses the thread until an operator approves or rejects
- Operators cannot decide their own request

Models and observability

- Cheap tier classifies, strong tier reasons; failover across 3 providers
- Every request writes ChatTrace + LlmCall (tokens, latency, USD cost)
- Evals: 41 cases, route / tool / trajectory / guardrail metrics, optional LLM judge
- Eval CI: keyless PR job, weekly live run with cost cap